

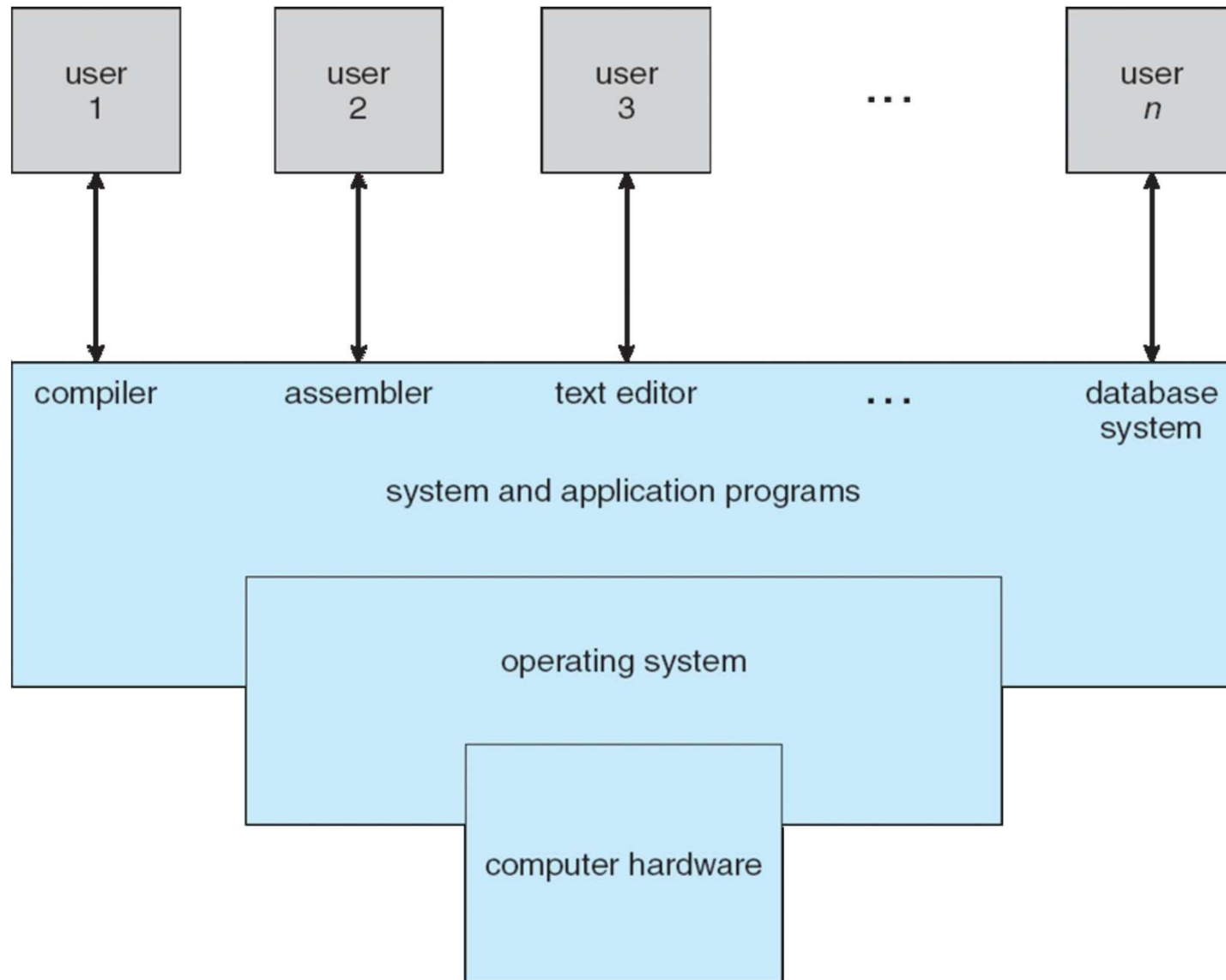
Operating Systems

CS2006

Lecture 2

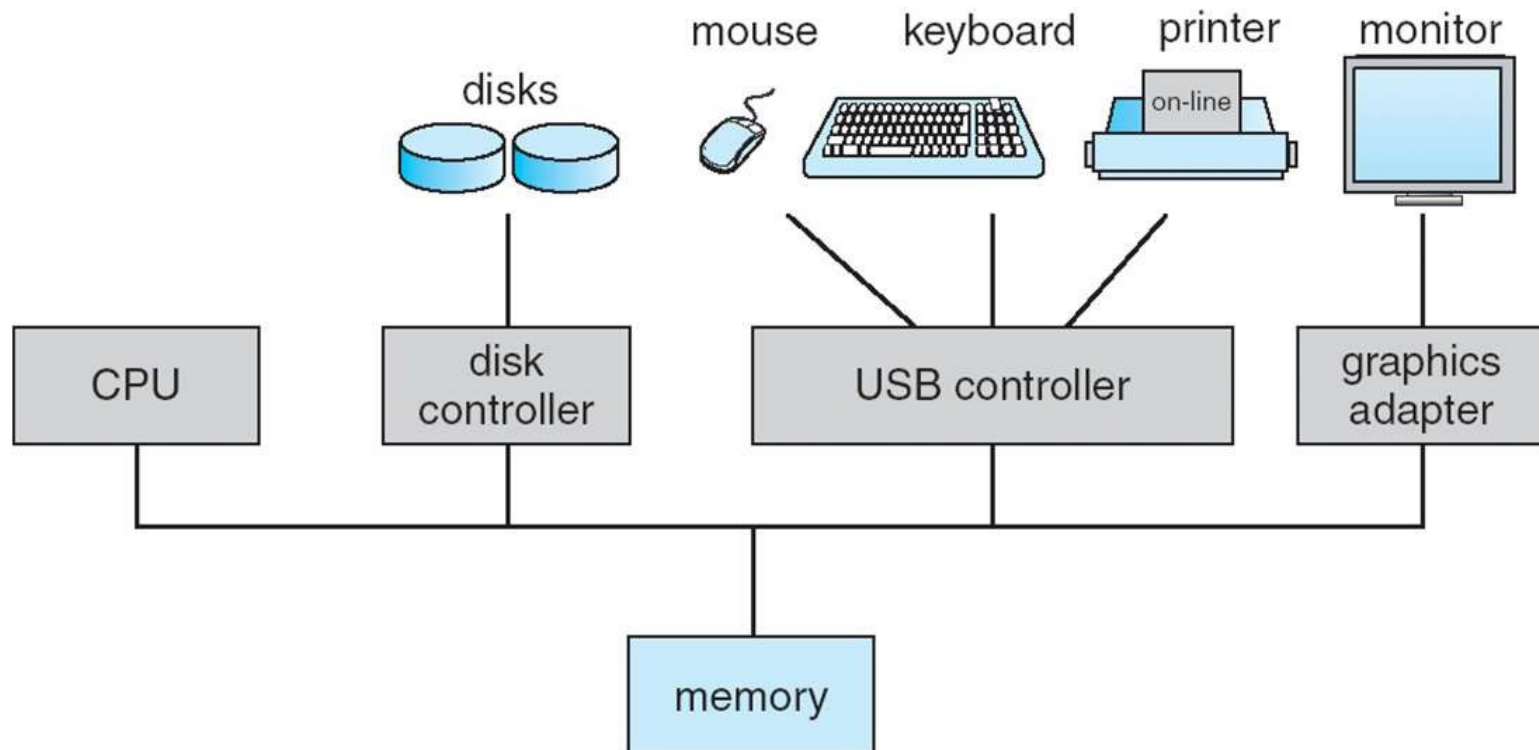
Basic Concepts

What is a Computer System?



Computer System Organization

- Computer-system operation
 - One or more CPUs, device controllers connect through common bus providing access to shared memory
 - Concurrent execution of CPUs and devices competing for memory cycles



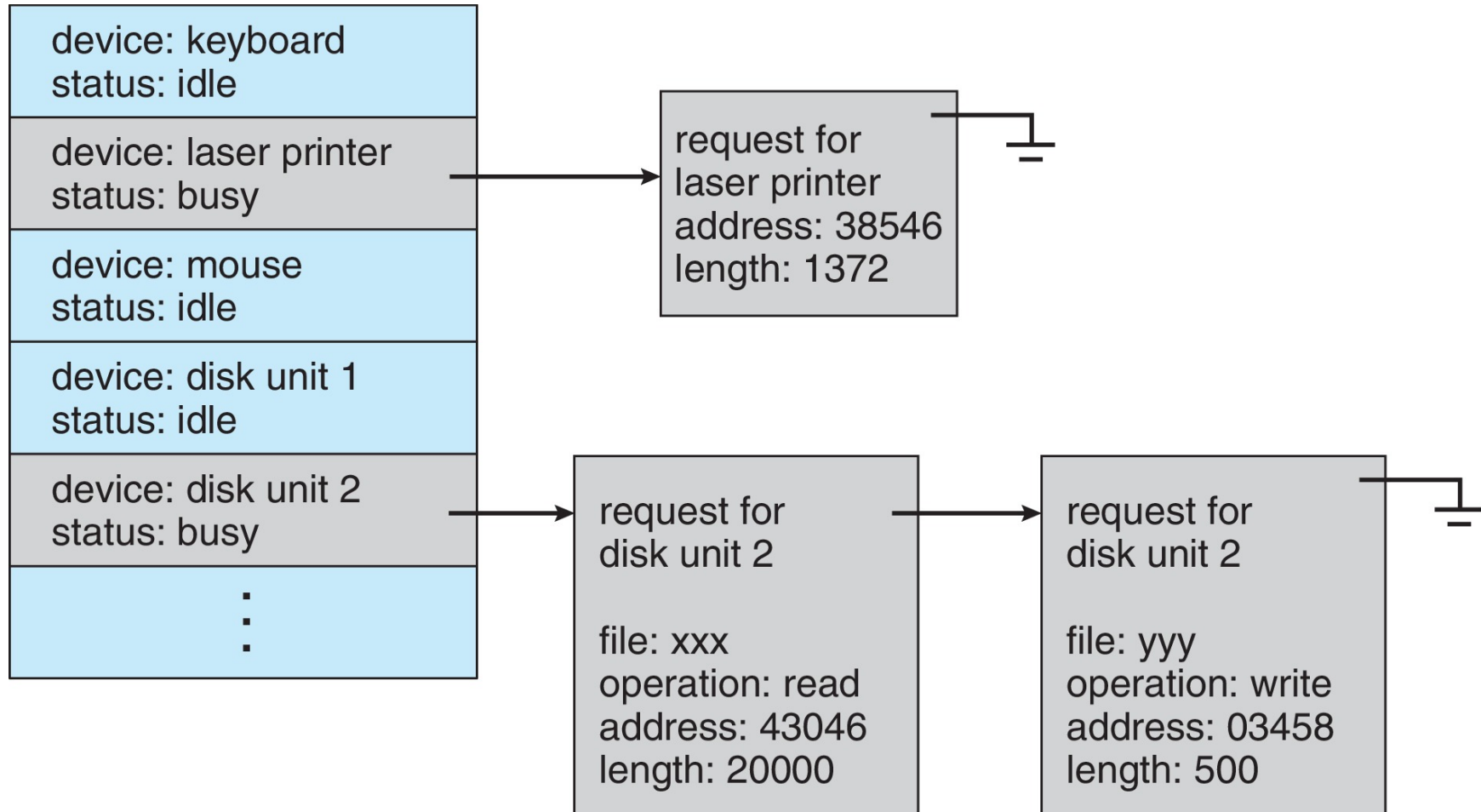
Device Controllers

- I/O devices generally consists of two parts:
 - An Electronic component: Device Controller
 - A Mechanical component: The device itself
- Each device controller is in charge of a specific device
- It maintains
 - Some local storage buffer
 - A set of special purpose register
- Accepts command from the Operating System
- For example to read data from the device.

Device Drivers

- Who would initiate the controller
- Who would determine read/write
- Who would tell the sector number
- A part of operating system called *Device Driver*
- Software or Hardware?
 - Software that talks to a controller
 - Gives it command
 - Accept Responses
- Same for all controllers?
- Different software for different type of controllers
- Each controller manufacturer has to supply a driver for each operating system it supports

Device Status Table



Operating System

- Modern Operating Systems are *Interrupt driven*
- If
 - No process to execute
 - No I/O device to service
 - No user to whom to respond
- Then
- OS will sit quietly, waiting for something to happen
- This something is *Interrupt*
- At a particular time either a user program is running or operating system is running

Interrupts

- **Interrupts** are signals by external devices, normally I/O devices. They tell the CPU to stop its current activities and execute the appropriate part of the operating system.

Types of Interrupts

- There are three (or broadly two) types of interrupts:
 - Hardware Interrupts
 - Software Interrupts
 - Traps

Hardware Interrupts

- **Hardware Interrupts** - generated by hardware devices to signal that they need some attention from the OS.
 - They may have just received some data (e.g., keystrokes on the keyboard or an data on the Ethernet card);
 - or they have just completed a task which the operating system previously requested, such as transferring data between the hard drive and memory.

Software Interrupts

- **Software Interrupts** are generated by programs when they want to request a system call to be performed by the operating system.

Traps

- **Traps** are generated by the CPU itself to indicate that some error or condition occurred for which assistance from the operating system is needed.

Interrupts and Traps

Occurrence	Cause	OS Mode
Trap	Unexpected occurrence within the program	Kernel
Interrupt	Event outside of program	Kernel
Subroutine call	Subroutine call branch in the program	User
System call	System call branch in the program	Kernel

Simple Instruction Cycle

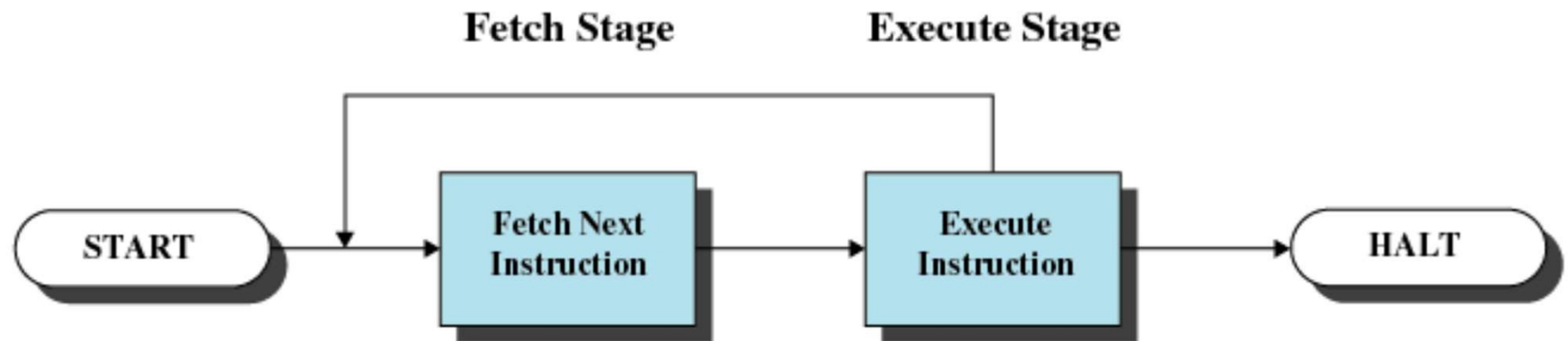


Figure 1.2 Basic Instruction Cycle

Interrupts

- Suspends the normal sequence of execution
- Used to improve processor utilization

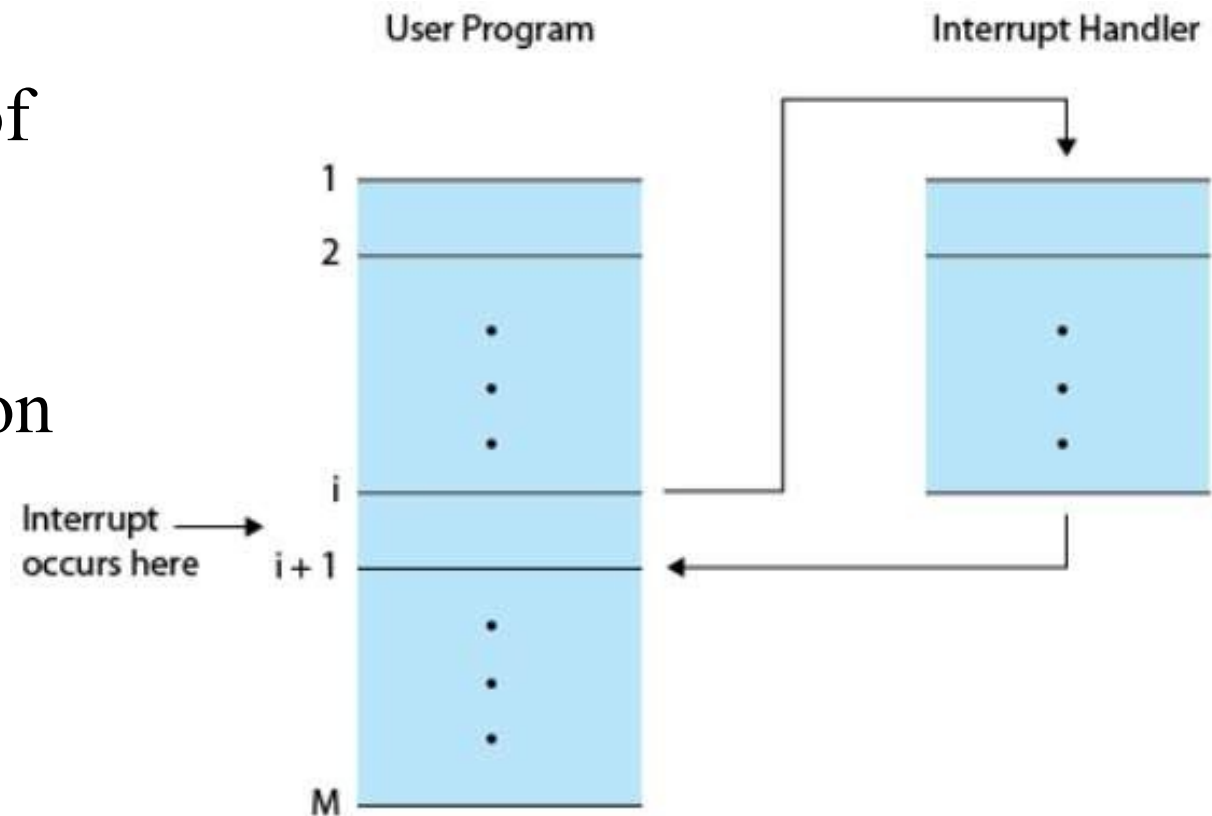


Figure 1.6 Transfer of Control via Interrupts

Interrupt Handling

- Different routines handle different type of interrupts.
- Called *Interrupt /ervice Routine (I/R)*
- When the CPU is interrupted it stops what it is doing
- The address of the interrupted instruction is saved
- A generic routine is run
 - This routine examines the nature of interrupt
 - Calls the corresponding ISR
- The ISR's are usually stored in the lower part of the memory
- After the interrupt is serviced, the saved address is loaded to the Program Counter
- The Interrupted computation can resume as though the interrupt had not occurred.

Interrupt Cycle

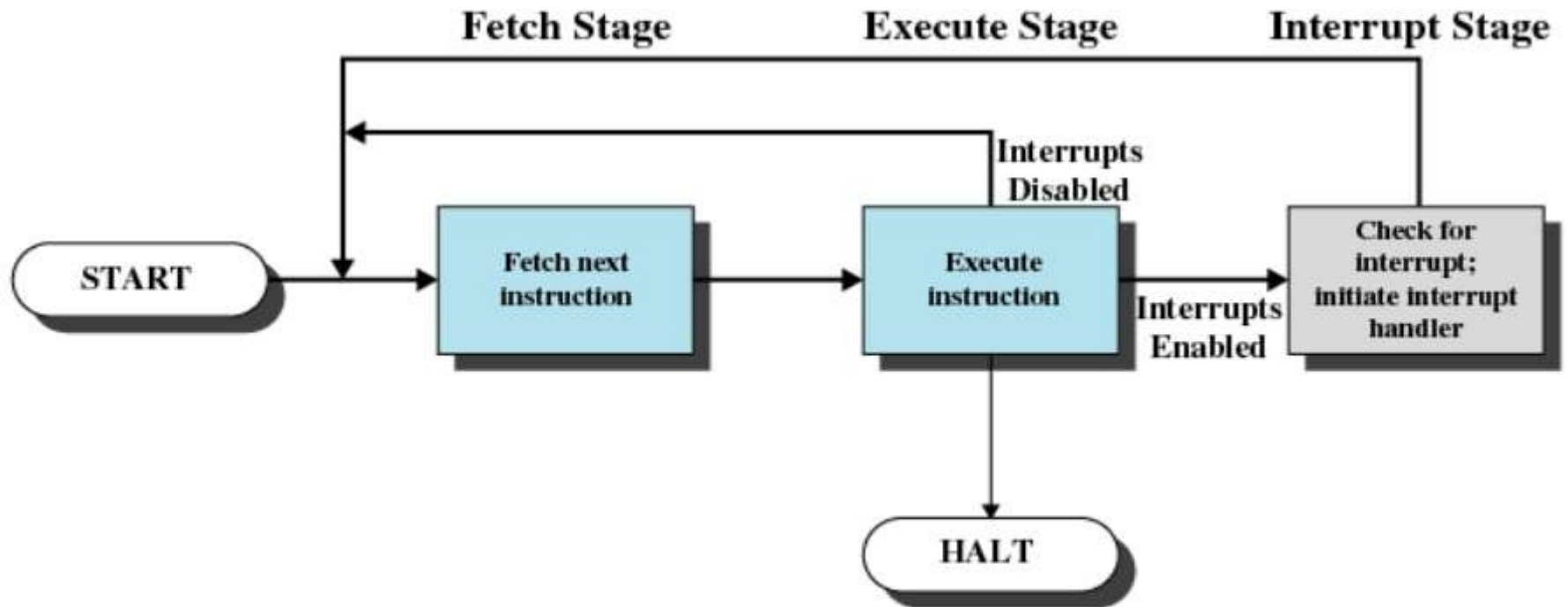
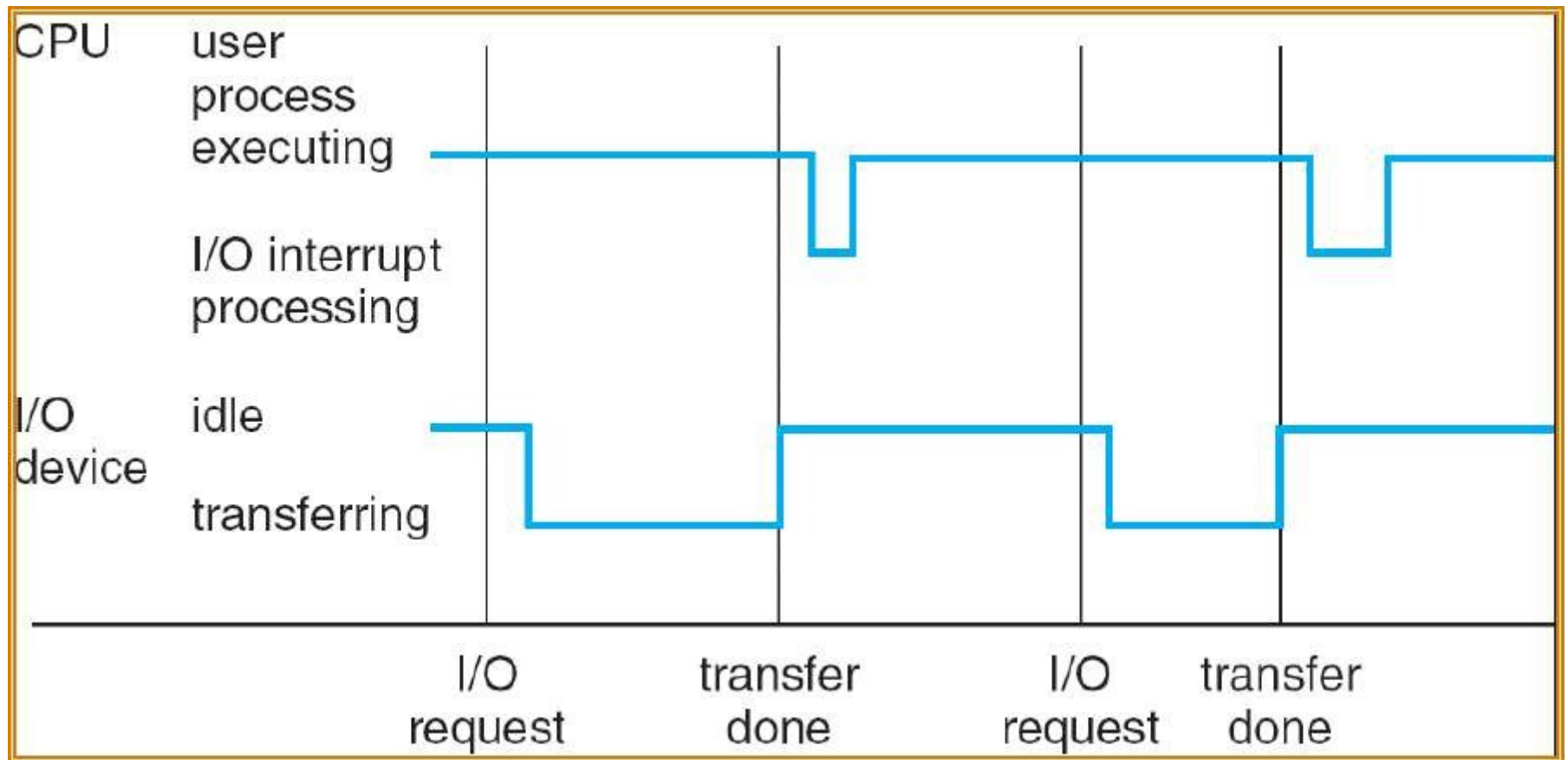


Figure 1.7 Instruction Cycle with Interrupts

Interrupt Timeline



Synchronous Interrupts

Synchronous Interrupts:

- 1. Happen at a specific time: When the CPU is executing a specific instruction.
- 2. Caused by the CPU: The CPU generates the interrupt.
- 3. Handled by the CPU: The CPU deals with the interrupt immediately.
- 4. Examples: Division by zero, page fault.

Asynchronous Interrupts

Asynchronous Interrupts:

- 1. Happen at any time: Can occur at any moment, not just when the CPU is executing a specific instruction.
- 2. Caused by external devices: Devices like keyboards, disks, or network interfaces generate the interrupt.
- 3. Handled by an interrupt controller: The interrupt controller sends a signal to the CPU to handle the interrupt.
- 4. Examples: Keyboard press, disk completion, network packet arrival.

Key Difference

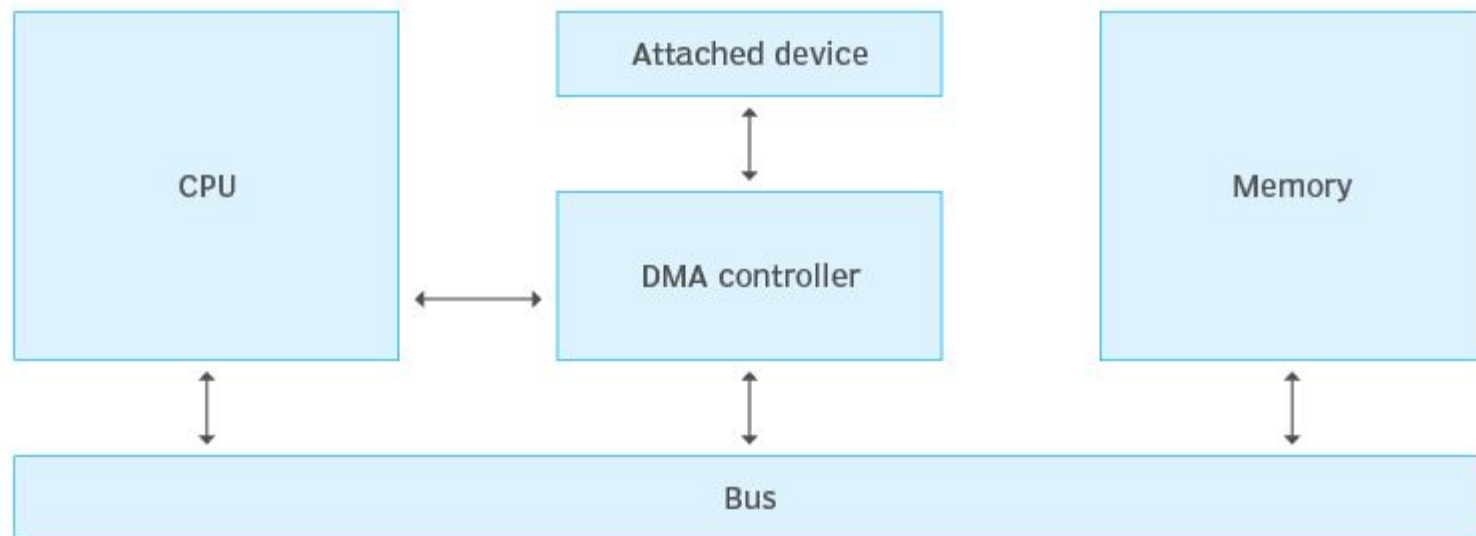
The key differences are:-

Timing: Synchronous interrupts happen at a specific time, while asynchronous interrupts can happen at any time.

Cause: Synchronous interrupts are caused by the CPU, while asynchronous interrupts are caused by external devices.

Direct Memory Access(DMA):

Where Direct Memory Access fits in



Storage Hierarchy:

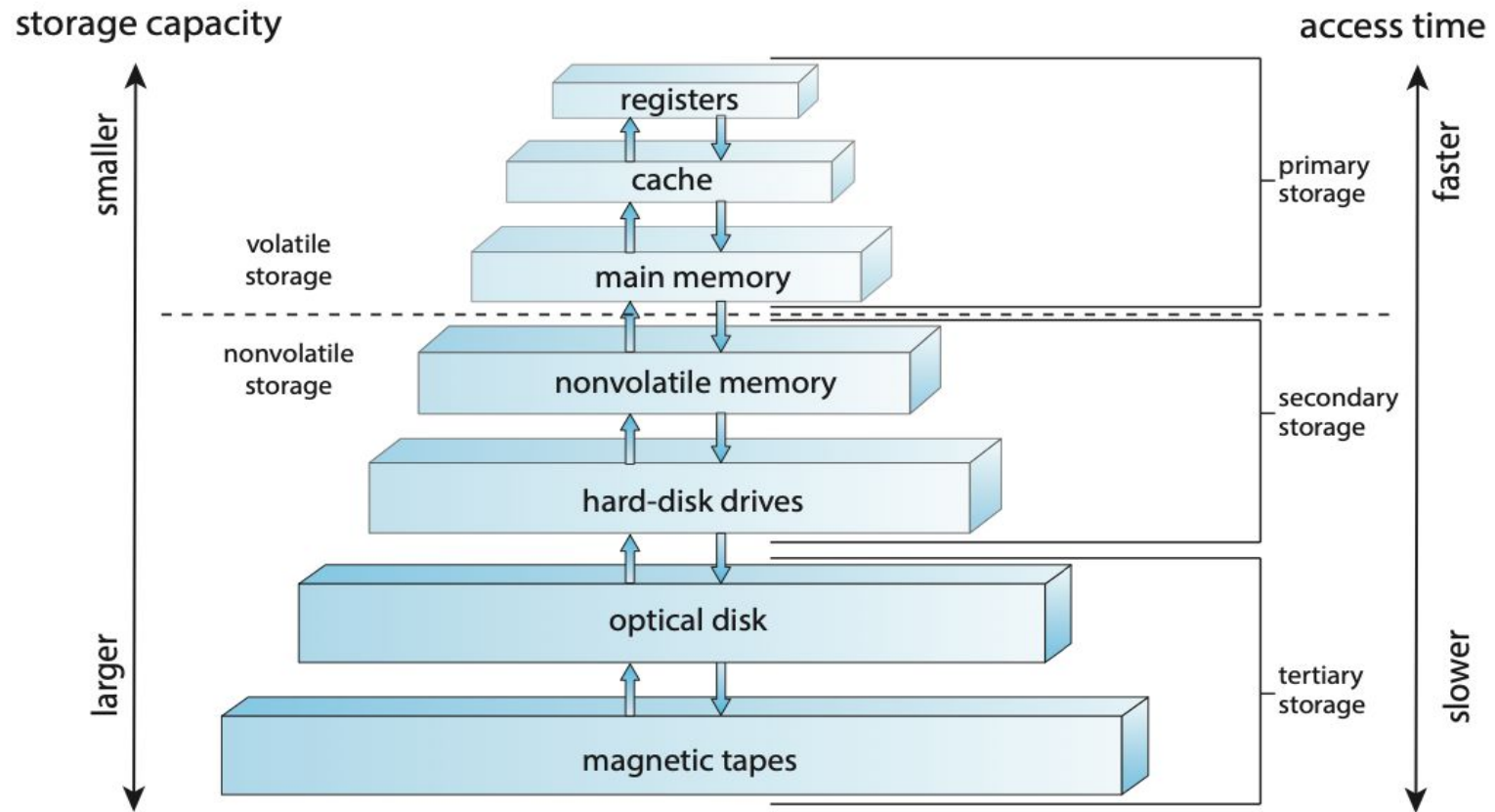


Figure 1.6 Storage-device hierarchy.